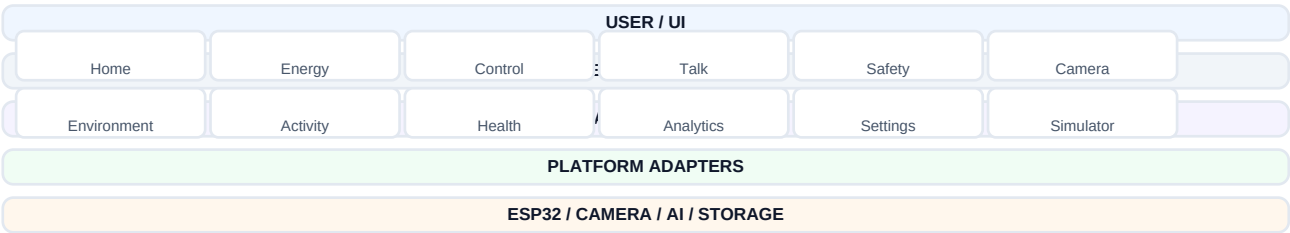


SOLAR ROBO

System Architecture & Module Implementation Specification

Precise build contract for AI coding models and human developers

Solar Robo — System Architecture



Purpose: define exact system boundaries, modules, files, functions, inputs, outputs, allowed dependencies, skeleton code and definition of done. A coding model should not need to infer missing responsibilities.

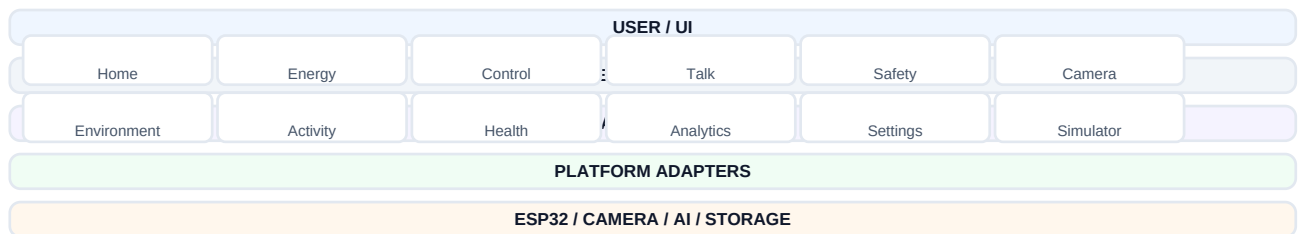
Core rule: Create the puzzle piece completely → verify it → mount it. A module must never expand its scope because an adjacent feature looks useful.

1. Architecture Rules

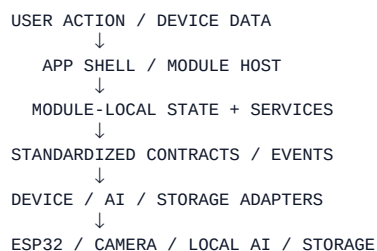
The application is a contract-driven modular client. Business modules are isolated; the App Shell composes them; adapters hide hardware/AI/storage details.

- Business modules must not import another business module.
- Cross-module data uses only `src/contracts/*.ts`.
- Device, AI and storage access uses `src/platform/*` interfaces.
- Every module has a mock path and can render without real hardware.
- UI components contain presentation logic only; domain/service decisions live in module services.
- AI can recommend; deterministic safety code decides whether a physical command is permitted.
- No unrequested feature is added. If a missing capability is discovered, the model reports it instead of implementing it.

Solar Robo — System Architecture



System flow



The same contract must be usable with a simulator and with production adapters. This is what makes module development independent.

2. Canonical Repository Structure

```
solar-robo-app/  
  ■■■ package.json  
  ■■■ tsconfig.json  
  ■■■ vite.config.ts  
  ■■■ index.html  
  ■■■ README.md  
  ■■■ src/  
    ■ ■■■ main.tsx  
    ■ ■■■ app/  
      ■ ■■■ App.tsx  
      ■ ■■■ routes.tsx  
      ■ ■■■ moduleRegistry.ts  
      ■ ■■■ ModuleHost.tsx  
      ■ ■■■ app.css  
      ■ ■■■ contracts/  
        ■ ■■■ index.ts  
        ■ ■■■ robo.ts  
        ■ ■■■ energy.ts  
        ■ ■■■ safety.ts  
        ■ ■■■ environment.ts  
        ■ ■■■ activity.ts  
        ■ ■■■ health.ts  
        ■ ■■■ notification.ts  
        ■ ■■■ conversation.ts  
        ■ ■■■ camera.ts  
        ■ ■■■ commands.ts  
      ■ ■■■ platform/  
        ■ ■■■ events/eventBus.ts  
        ■ ■■■ events/eventTypes.ts  
        ■ ■■■ device/DeviceAdapter.ts  
        ■ ■■■ device/MockDeviceAdapter.ts  
        ■ ■■■ ai/AIAdapter.ts  
        ■ ■■■ ai/MockAIAdapter.ts  
        ■ ■■■ storage/StorageAdapter.ts  
        ■ ■■■ storage/MemoryStorageAdapter.ts  
        ■ ■■■ clock/Clock.ts  
      ■ ■■■ modules/  
        ■ ■■■ home/  
        ■ ■■■ onboarding/  
        ■ ■■■ energy/  
        ■ ■■■ control/  
        ■ ■■■ talk/  
        ■ ■■■ safety/  
        ■ ■■■ camera/  
        ■ ■■■ environment/  
        ■ ■■■ activity/  
        ■ ■■■ health/  
        ■ ■■■ analytics/  
        ■ ■■■ notifications/  
        ■ ■■■ settings/  
        ■ ■■■ simulator/  
      ■ ■■■ shared/  
        ■ ■■■ components/  
        ■ ■■■ hooks/  
        ■ ■■■ formatting/  
    ■■■ tests/  
      ■■■ contracts/  
      ■■■ integration/
```

Standard module template

```
src/modules/<module>/  
  ■■■ index.ts           # only public entry point  
  ■■■ types.ts          # module-local types  
  ■■■ state/<module>Store.ts  
  ■■■ services/<module>Service.ts  
  ■■■ ui/<Module>Page.tsx  
  ■■■ ui/components/<Module>*.tsx  
  ■■■ mock/<module>Mock.ts  
  ■■■ tests/<module>Store.test.ts  
  ■■■ tests/<module>Service.test.ts  
  ■■■ README.md
```

A module may omit a file that is genuinely unnecessary, but it may not create a second architectural location for the same responsibility.

3. Canonical Cross-Module Contracts

Contract files contain types only. They do not import React, ESP32 SDKs, LLM SDKs, database clients, or module internals.

```
export type RoboMode =
  | "normal" | "optimizing" | "conserving"
  | "protecting" | "fault" | "safe";

export interface RoboSnapshot {
  deviceId: string; name: string; connected: boolean;
  mode: RoboMode; panelAngleDeg: number; targetAngleDeg: number;
  generationW: number; batteryPct: number; lastUpdatedAt: string;
}

export interface EnergySnapshot {
  generatedW: number; consumedW: number; batteryPct: number;
  batteryPowerW: number; reservePct: number;
  gridState: "importing" | "export-ready" | "isolated" | "unknown";
  updatedAt: string;
}

export type RoboCommand =
  | { type: "MOVE_TO_ANGLE"; angleDeg: number }
  | { type: "STOP_MOTION" }
  | { type: "SAFE_POSITION" };

export interface CommandResult {
  accepted: boolean; commandId: string; reason?: string; executedAt?: string;
}

export type SafetyLevel = "normal" | "caution" | "protecting" | "fault" | "emergency";
export interface SafetyEvent {
  id: string; level: SafetyLevel; code: string; message: string;
  createdAt: string; acknowledged: boolean;
}

export interface EnvironmentSnapshot {
  temperatureC: number; humidityPct: number; lightLux: number;
  windMps: number; rainDetected: boolean; panelTemperatureC: number;
  updatedAt: string;
}

export interface ActivityEvent {
  id: string; type: "movement"|"energy"|"safety"|"system"|"ai"|"recovery";
  title: string; detail: string; timestamp: string;
}
```

Contract rule

If two modules need to exchange data, add or reuse a shared contract. Never solve the problem by importing the other module's service/store.

4. Platform Interfaces

```
export interface DeviceAdapter {
  getRoboSnapshot(): Promise<RoboSnapshot>;
  getEnergySnapshot(): Promise<EnergySnapshot>;
  getEnvironmentSnapshot(): Promise<EnvironmentSnapshot>;
  sendCommand(command: RoboCommand): Promise<CommandResult>;
  subscribeSnapshot(listener: (s: RoboSnapshot) => void): () => void;
}

export interface AIRequest {
  message: string;
  context: Record<string, unknown>;
}
export interface AIResponse {
  text: string; intent?: string; confidence?: number;
}
export interface AIAdapter {
  respond(request: AIRequest): Promise<AIResponse>;
}

export interface StorageAdapter {
  get<T>(key: string): Promise<T | null>;
  set<T>(key: string, value: T): Promise<void>;
  remove(key: string): Promise<void>;
}
```

Adapter I/O

Function	Input	Output	Boundary
getRoboSnapshot	none	Promise<RoboSnapshot>	Read only
getEnergySnapshot	none	Promise<EnergySnapshot>	Read only
getEnvironmentSnapshot	none	Promise<EnvironmentSnapshot>	Read only
sendCommand	RoboCommand	Promise<CommandResult>	Must pass safety gate before physical execution
AIAdapter.respond	AIRequest	Promise<AIResponse>	Model invocation only
Storage.get/set/remove	key/value	Promise	Persistence only

5. Device Onboarding

Adds a Robo and verifies connection; it never operates the panel.

Required folder/files

```
src/modules/onboarding/index.ts
src/modules/onboarding/types.ts
src/modules/onboarding/state/onboardingStore.ts
src/modules/onboarding/services/onboardingService.ts
src/modules/onboarding/ui/OnboardingPage.tsx
src/modules/onboarding/ui/components/DeviceCard.tsx
src/modules/onboarding/mock/onboardingMock.ts
src/modules/onboarding/tests/onboardingService.test.ts
src/modules/onboarding/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	DeviceCandidate[]	DeviceAdapter/mock	Consume only through declared boundary
INPUT	User DeviceConfig	StorageAdapter	Consume only through declared boundary
OUTPUT	O	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	b	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	g	module public boundary	No undocumented side effects
OUTPUT	R	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	u	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	D	module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	V	module public boundary	No undocumented side effects
OUTPUT	I	module public boundary	No undocumented side effects
OUTPUT	C	module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	–	module public boundary	No undocumented side effects
OUTPUT	A	module public boundary	No undocumented side effects
OUTPUT	D	module public boundary	No undocumented side effects
OUTPUT	D	module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects

Direction	Input / Output	Source / consumer	Rule
OUTPUT	D	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
scanDevices()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
connectDevice(id)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
saveConfiguration(config)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
completeOnboarding(config)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: DeviceAdapter/mock
- Read/use: StorageAdapter

Must not implement: do not control angle; do not optimize energy; do not call AI

6. Home Command Center

First-glance status presentation assembled from standardized snapshots.

Required folder/files

```
src/modules/home/index.ts
src/modules/home/types.ts
src/modules/home/state/homeStore.ts
src/modules/home/services/homeService.ts
src/modules/home/ui/HomePage.tsx
src/modules/home/ui/components/RoboStatusCard.tsx
src/modules/home/ui/components/EnergySummaryCard.tsx
src/modules/home/ui/components/AlertSummaryCard.tsx
src/modules/home/mock/homeMock.ts
src/modules/home/tests/homeService.test.ts
src/modules/home/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	RoboSnapshot	Device/contract data	Consume only through declared boundary
INPUT	EnergySnapshot	event/storage data	Consume only through declared boundary
INPUT	SafetyEvent[]	Device/contract data	Consume only through declared boundary
OUTPUT	H	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	V	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	w	module public boundary	No undocumented side effects
OUTPUT	M	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	g	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects

Direction	Input / Output	Source / consumer	Rule
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
loadHomeSnapshot()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
formatRoboStatus(snapshot)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
getActiveAlertSummary(events)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: Device/contract data
- Read/use: event/storage data

Must not implement: do not own source data; do not control hardware; do not call AI

7. Energy Center

Generation, consumption, battery, reserve, grid-aware state and history presentation.

Required folder/files

```
src/modules/energy/index.ts
src/modules/energy/types.ts
src/modules/energy/state/energyStore.ts
src/modules/energy/services/energyService.ts
src/modules/energy/ui/EnergyPage.tsx
src/modules/energy/ui/components/EnergyFlowCard.tsx
src/modules/energy/ui/components/BatteryCard.tsx
src/modules/energy/ui/components/EnergyHistoryChart.tsx
src/modules/energy/mock/energyMock.ts
src/modules/energy/tests/energyService.test.ts
src/modules/energy/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	EnergySnapshot	DeviceAdapter	Consume only through declared boundary
INPUT	EnergyHistoryPoint[]	StorageAdapter/mock	Consume only through declared boundary
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	g	module public boundary	No undocumented side effects
OUTPUT	y	module public boundary	No undocumented side effects
OUTPUT	V	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	w	module public boundary	No undocumented side effects
OUTPUT	M	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	R	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	C	module public boundary	No undocumented side effects
OUTPUT	h	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects

Direction	Input / Output	Source / consumer	Rule
OUTPUT	g	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	R	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	q	module public boundary	No undocumented side effects
OUTPUT	u	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
loadEnergySnapshot()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
loadHistory(range)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
buildEnergyViewModel(snapshot,history)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: DeviceAdapter
- Read/use: StorageAdapter/mock

Must not implement: do not move panel; do not enforce hardware limits; do not claim certified grid export

8. Robo Control

Manual/auto UI and typed movement commands.

Required folder/files

```
src/modules/control/index.ts
src/modules/control/types.ts
src/modules/control/state/controlStore.ts
src/modules/control/services/controlService.ts
src/modules/control/ui/ControlPage.tsx
src/modules/control/ui/components/AngleControl.tsx
src/modules/control/ui/components/ModeToggle.tsx
src/modules/control/ui/components/MotionStatus.tsx
src/modules/control/mock/controlMock.ts
src/modules/control/tests/controlService.test.ts
src/modules/control/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	RoboSnapshot	DeviceAdapter	Consume only through declared boundary
INPUT	user angle request	Safety command boundary	Consume only through declared boundary
OUTPUT	R	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	b	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	C	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	C	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects
OUTPUT	R	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	u	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects

Direction	Input / Output	Source / consumer	Rule
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	C	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects
OUTPUT	S	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
validateAngleInput(angle)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
createMoveCommand(angle)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
sendMoveCommand(angle)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
stopMotion()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
requestSafePosition()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: DeviceAdapter
- Read/use: Safety command boundary

Must not implement: do not bypass safety; do not call AI; do not access GPIO directly

9. Robo Talk

Chat/voice presentation and AI request/response handling; no physical actuation.

Required folder/files

```
src/modules/talk/index.ts
src/modules/talk/types.ts
src/modules/talk/state/talkStore.ts
src/modules/talk/services/talkService.ts
src/modules/talk/ui/TalkPage.tsx
src/modules/talk/ui/components/MessageList.tsx
src/modules/talk/ui/components/Composer.tsx
src/modules/talk/ui/components/VoiceState.tsx
src/modules/talk/mock/talkMock.ts
src/modules/talk/tests/talkService.test.ts
src/modules/talk/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	User message	AIAdapter	Consume only through declared boundary
INPUT	ConversationMessage[]	StorageAdapter/mock	Consume only through declared boundary
OUTPUT	C	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	M	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	g	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	C	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects

Direction	Input / Output	Source / consumer	Rule
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
buildAIRequest(message,history)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
sendMessage(message)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
setVoiceState(state)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: AIAdapter
- Read/use: StorageAdapter/mock

Must not implement: do not send device commands; do not implement STT/TTS engines; do not change safety

10. Safety & Emergency

Deterministic safety gate, emergency stop, safe position and incident UX.

Required folder/files

```
src/modules/safety/index.ts
src/modules/safety/types.ts
src/modules/safety/state/safetyStore.ts
src/modules/safety/services/safetyPolicy.ts
src/modules/safety/services/safetyService.ts
src/modules/safety/ui/SafetyPage.tsx
src/modules/safety/ui/components/SafetyBanner.tsx
src/modules/safety/ui/components/EmergencyStopButton.tsx
src/modules/safety/ui/components/IncidentList.tsx
src/modules/safety/mock/safetyMock.ts
src/modules/safety/tests/safetyPolicy.test.ts
src/modules/safety/tests/safetyService.test.ts
src/modules/safety/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	RoboCommand	DeviceAdapter	Consume only through declared boundary
INPUT	EnvironmentSnapshot	contracts	Consume only through declared boundary
INPUT	Device health/fault state	event bus	Consume only through declared boundary
OUTPUT	S	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	f	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	y	module public boundary	No undocumented side effects
OUTPUT	D	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	c	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	S	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	f	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	y	module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects

Direction	Input / Output	Source / consumer	Rule
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	f	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	c	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
evaluateCommand(command,context)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
triggerEmergencyStop(reason)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
requestSafePosition(reason)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
acknowledgeIncident(id)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: DeviceAdapter
- Read/use: contracts
- Read/use: event bus

Must not implement: do not use LLM for safety; do not silently change thresholds; do not duplicate telemetry

11. Live Camera

Camera connection, live frame and snapshot presentation; no vision reasoning.

Required folder/files

```
src/modules/camera/index.ts
src/modules/camera/types.ts
src/modules/camera/state/cameraStore.ts
src/modules/camera/services/cameraService.ts
src/modules/camera/ui/CameraPage.tsx
src/modules/camera/ui/components/CameraViewport.tsx
src/modules/camera/ui/components/CameraStatus.tsx
src/modules/camera/mock/cameraMock.ts
src/modules/camera/tests/cameraService.test.ts
src/modules/camera/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	Camera state/frame	Camera adapter/mock	Consume only through declared boundary
INPUT	user camera action	Camera adapter/mock	Consume only through declared boundary
OUTPUT	C	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	S	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	f	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	/	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	p	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	h	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	b	module public boundary	No undocumented side effects

Direction	Input / Output	Source / consumer	Rule
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	u	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	y	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
startCamera()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
stopCamera()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
captureSnapshot()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
buildCameraState()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: Camera adapter/mock

Must not implement: do not face-recognize; do not VLM-reason; do not store continuous video by default

12. Environment

Current local environmental measurements and bounded condition presentation.

Required folder/files

```
src/modules/environment/index.ts
src/modules/environment/types.ts
src/modules/environment/state/environmentStore.ts
src/modules/environment/services/environmentService.ts
src/modules/environment/ui/EnvironmentPage.tsx
src/modules/environment/ui/components/ConditionCard.tsx
src/modules/environment/ui/components/WeatherRiskCard.tsx
src/modules/environment/mock/environmentMock.ts
src/modules/environment/tests/environmentService.test.ts
src/modules/environment/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	EnvironmentSnapshot	DeviceAdapter/simulator	Consume only through declared boundary
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	V	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	w	module public boundary	No undocumented side effects
OUTPUT	M	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	p	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects

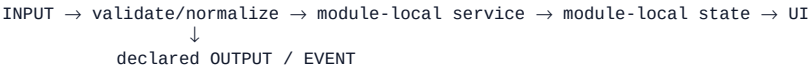
Direction	Input / Output	Source / consumer	Rule
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
loadEnvironment()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
buildEnvironmentViewModel(snapshot)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
detectLocalConditionChange(previous,current)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence



- Read/use: DeviceAdapter/simulator

Must not implement: do not forecast weather; do not move panel; do not create safety thresholds in UI

13. Activity / Robo Memory

Chronological important-event history.

Required folder/files

```
src/modules/activity/index.ts
src/modules/activity/types.ts
src/modules/activity/state/activityStore.ts
src/modules/activity/services/activityService.ts
src/modules/activity/ui/ActivityPage.tsx
src/modules/activity/ui/components/ActivityItem.tsx
src/modules/activity/mock/activityMock.ts
src/modules/activity/tests/activityService.test.ts
src/modules/activity/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	ActivityEvent	EventBus	Consume only through declared boundary
INPUT	DateRange/filter	StorageAdapter	Consume only through declared boundary
OUTPUT	A	module public boundary	No undocumented side effects
OUTPUT	c	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	y	module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	[	module public boundary	No undocumented side effects
OUTPUT	]	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	/	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
appendEvent(event)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Function	Input	Output	Exact task
queryEvents(filter)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
summarizeEvent(event)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: EventBus
- Read/use: StorageAdapter

Must not implement: do not log every raw sensor sample; do not own analytics

14. Health & Diagnostics

Shows controller, motor, battery, sensor, camera and connection health.

Required folder/files

```
src/modules/health/index.ts
src/modules/health/types.ts
src/modules/health/state/healthStore.ts
src/modules/health/services/healthService.ts
src/modules/health/ui/HealthPage.tsx
src/modules/health/ui/components/HealthCheckItem.tsx
src/modules/health/mock/healthMock.ts
src/modules/health/tests/healthService.test.ts
src/modules/health/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	DeviceHealth	DeviceAdapter	Consume only through declared boundary
INPUT	diagnostic records	StorageAdapter	Consume only through declared boundary
OUTPUT	H	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	h	module public boundary	No undocumented side effects
OUTPUT	V	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	w	module public boundary	No undocumented side effects
OUTPUT	M	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	H	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	h	module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
loadHealth()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
buildHealthViewModel(health)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
findCriticalIssues(health)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: DeviceAdapter
- Read/use: StorageAdapter

Must not implement: do not repair hardware; do not alter motor parameters; do not duplicate safety policy

15. Analytics & Performance

Historical energy/activity calculations and charts.

Required folder/files

```
src/modules/analytics/index.ts
src/modules/analytics/types.ts
src/modules/analytics/state/analyticsStore.ts
src/modules/analytics/services/analyticsService.ts
src/modules/analytics/ui/AnalyticsPage.tsx
src/modules/analytics/ui/components/MetricCard.tsx
src/modules/analytics/ui/components/PerformanceChart.tsx
src/modules/analytics/mock/analyticsMock.ts
src/modules/analytics/tests/analyticsService.test.ts
src/modules/analytics/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	EnergyHistoryPoint[]	StorageAdapter	Consume only through declared boundary
INPUT	ActivityEvent[]	StorageAdapter	Consume only through declared boundary
OUTPUT	A	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	l	module public boundary	No undocumented side effects
OUTPUT	y	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	c	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	R	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	p	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
calculateEnergyTotal(points)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
calculateBatteryUtilization(points)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
buildPerformanceReport(energy,activity)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: StorageAdapter

Must not implement: do not fabricate missing data; do not control device; do not claim comparison results without evidence

16. Notifications

Presentation/inbox for standardized user-facing events.

Required folder/files

```
src/modules/notifications/index.ts
src/modules/notifications/types.ts
src/modules/notifications/state/notificationStore.ts
src/modules/notifications/services/notificationService.ts
src/modules/notifications/ui/NotificationsPage.tsx
src/modules/notifications/ui/components/NotificationItem.tsx
src/modules/notifications/mock/notificationMock.ts
src/modules/notifications/tests/notificationService.test.ts
src/modules/notifications/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	NotificationEvent	EventBus	Consume only through declared boundary
INPUT	user read action	StorageAdapter	Consume only through declared boundary
OUTPUT	N	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	f	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	c	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	[	module public boundary	No undocumented side effects
OUTPUT	]	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
ingest(event)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
markRead(id)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
getUnreadCount()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: EventBus
- Read/use: StorageAdapter

Must not implement: do not decide root cause; do not repeatedly notify same unresolved event

17. Settings & Configuration

User preferences; safety-critical hardware limits remain outside editable settings.

Required folder/files

```
src/modules/settings/index.ts
src/modules/settings/types.ts
src/modules/settings/state/settingsStore.ts
src/modules/settings/services/settingsService.ts
src/modules/settings/ui/SettingsPage.tsx
src/modules/settings/ui/components/GeneralSettings.tsx
src/modules/settings/ui/components/NotificationSettings.tsx
src/modules/settings/ui/components/VoiceSettings.tsx
src/modules/settings/mock/settingsMock.ts
src/modules/settings/tests/settingsService.test.ts
src/modules/settings/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	PersistedSettings	StorageAdapter	Consume only through declared boundary
INPUT	user patch	StorageAdapter	Consume only through declared boundary
OUTPUT	S	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	g	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	C	module public boundary	No undocumented side effects
OUTPUT	h	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	g	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	p	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects

Direction	Input / Output	Source / consumer	Rule
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	d	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	g	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
loadSettings()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
updateSettings(patch)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
resetSettings()	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: StorageAdapter

Must not implement: do not edit hidden safety thresholds; do not access GPIO; do not implement auth

18. Developer Simulator

Deterministic fake world producing the same contracts as real adapters.

Required folder/files

```
src/modules/simulator/index.ts
src/modules/simulator/types.ts
src/modules/simulator/state/simulatorStore.ts
src/modules/simulator/services/simulatorEngine.ts
src/modules/simulator/ui/SimulatorPage.tsx
src/modules/simulator/ui/components/SimulationControl.tsx
src/modules/simulator/mock/scenarios.ts
src/modules/simulator/tests/simulatorEngine.test.ts
src/modules/simulator/README.md
```

Exact module I/O

Direction	Input / Output	Source / consumer	Rule
INPUT	ScenarioConfig	Clock	Consume only through declared boundary
INPUT	tick(dt)	EventBus	Consume only through declared boundary
OUTPUT	R	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	b	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	S	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	p	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	h	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	g	module public boundary	No undocumented side effects
OUTPUT	y	module public boundary	No undocumented side effects
OUTPUT	S	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	p	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	h	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects

Direction	Input / Output	Source / consumer	Rule
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	i	module public boundary	No undocumented side effects
OUTPUT	r	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	m	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	S	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	p	module public boundary	No undocumented side effects
OUTPUT	s	module public boundary	No undocumented side effects
OUTPUT	h	module public boundary	No undocumented side effects
OUTPUT	o	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	+	module public boundary	No undocumented side effects
OUTPUT		module public boundary	No undocumented side effects
OUTPUT	S	module public boundary	No undocumented side effects
OUTPUT	a	module public boundary	No undocumented side effects
OUTPUT	f	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects
OUTPUT	y	module public boundary	No undocumented side effects
OUTPUT	E	module public boundary	No undocumented side effects
OUTPUT	v	module public boundary	No undocumented side effects
OUTPUT	e	module public boundary	No undocumented side effects
OUTPUT	n	module public boundary	No undocumented side effects
OUTPUT	t	module public boundary	No undocumented side effects

Functions

Function	Input	Output	Exact task
applyScenario(scenario)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
tick(dt)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
buildSnapshots(state)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic
injectFault(type)	declared contract / user data	declared result	Implement only the named responsibility; no cross-module business logic

Allowed resources

Execution sequence

INPUT → validate/normalize → module-local service → module-local state → UI
↓
declared OUTPUT / EVENT

- Read/use: Clock
- Read/use: EventBus
- Read/use: contracts

Must not implement: do not connect real hardware; do not become production adapter; no hidden randomness in tests

19. Exact File Assignment for Key Modules

The following filenames are fixed for the first implementation pass. A coding model should not rename them for stylistic reasons.

Module	Exact files
App Shell	src/app/App.tsx; routes.tsx; moduleRegistry.ts; ModuleHost.tsx; app.css
Contracts	src/contracts/robo.ts; energy.ts; safety.ts; environment.ts; activity.ts; health.ts; notification.ts; conversation.ts; camera.ts; commands.ts
Home	src/modules/home/index.ts; types.ts; state/homeStore.ts; services/homeService.ts; ui/HomePage.tsx; ui/components/RoboStatusCard.tsx; EnergySummaryC
Energy	src/modules/energy/index.ts; types.ts; state/energyStore.ts; services/energyService.ts; ui/EnergyPage.tsx; ui/components/EnergyFlowCard.tsx; BatteryCard.t
Control	src/modules/control/index.ts; types.ts; state/controlStore.ts; services/controlService.ts; ui/ControlPage.tsx; ui/components/AngleControl.tsx; ModeToggle.tsx;
Safety	src/modules/safety/index.ts; types.ts; state/safetyStore.ts; services/safetyPolicy.ts; services/safetyService.ts; ui/SafetyPage.tsx; ui/components/SafetyBanner
Simulator	src/modules/simulator/index.ts; types.ts; state/simulatorStore.ts; services/simulatorEngine.ts; ui/SimulatorPage.tsx; ui/components/SimulationControl.tsx; mo

Naming rule

Component filenames are PascalCase; stores/services/mocks are camelCase; contract files are lowercase domain names. Avoid generic names such as utils.ts, manager.ts or helpers.ts when a domain-specific filename is possible.

20. Skeleton Code Patterns

These snippets define the expected coding style: typed boundaries, small functions, explicit dependencies, no hidden global behavior.

```
export interface ModuleDependencies {
  device?: DeviceAdapter;
  ai?: AIAdapter;
  storage?: StorageAdapter;
  eventBus?: EventBus;
}

export function createEnergyService(deps: ModuleDependencies) {
  async function loadEnergySnapshot(): Promise<EnergySnapshot> {
    if (!deps.device) throw new Error("DeviceAdapter is required");
    return deps.device.getEnergySnapshot();
  }

  function buildEnergyViewModel(s: EnergySnapshot): EnergyViewModel {
    return {
      generatedWLabel: `${Math.round(s.generatedW)} W`,
      consumedWLabel: `${Math.round(s.consumedW)} W`,
      batteryLabel: `${Math.round(s.batteryPct)}%`,
      reserveLabel: `${Math.round(s.reservePct)}%`,
      gridStateLabel: s.gridState,
    };
  }

  return { loadEnergySnapshot, buildEnergyViewModel };
}

export interface CommandGateway {
  execute(command: RoboCommand): Promise<CommandResult>;
}

export async function sendMoveCommand(
  angleDeg: number,
  deps: { gateway: CommandGateway }
): Promise<CommandResult> {
  const command: RoboCommand = { type: "MOVE_TO_ANGLE", angleDeg };
  return deps.gateway.execute(command);
}
```

Important: the Control service above does not itself claim the command is safe. In the production wiring, the command path must be Safety → DeviceAdapter, or an equivalent single safety-gated command gateway. The exact wiring should be finalized before real hardware control is enabled.

```
export function createMockDevice(): DeviceAdapter {
  return {
    async getRoboSnapshot() { return mockRoboSnapshot; },
    async getEnergySnapshot() { return mockEnergySnapshot; },
    async getEnvironmentSnapshot() { return mockEnvironmentSnapshot; },
    async sendCommand(command) {
      return { accepted: true, commandId: crypto.randomUUID() };
    },
    subscribeSnapshot(listener) {
      listener(mockRoboSnapshot);
      return () => {};
    },
  };
}
```

21. AI Coding Model Work Protocol

Use this exact instruction block when delegating a module to an AI coding model.

YOU ARE IMPLEMENTING ONLY: <MODULE>

READ FIRST

- This specification
- The exact module section
- Existing contracts in src/contracts
- Existing platform interfaces in src/platform

BEFORE CODING

1. Inspect the repository.
2. Confirm the required files already do not conflict.
3. Reuse existing shared components.
4. Do not invent a new architecture.

IMPLEMENT

1. Create only the listed module files.
2. Implement only listed functions.
3. Use exact contract names and shapes.
4. Use mocks if infrastructure is unavailable.
5. Keep state inside the module.
6. Keep hardware/AI/storage behind adapters.

FORBIDDEN

- Importing another business module.
- Adding unrequested features.
- Modifying global architecture without approval.
- Creating duplicate contracts.
- Direct GPIO/device SDK calls from UI.
- LLM decisions for deterministic safety.

VERIFY

- Typecheck/build.
- Run module tests.
- Test success + failure + offline/empty states.
- Search for forbidden imports.
- Check every required file exists.

FINAL REPORT

Files created/changed:

Functions implemented:

Inputs consumed:

Outputs produced:

Tests passed:

Scope additions: NONE unless explicitly approved.

22. Integration & Definition of Done

Integration is a mechanical mounting operation, not a second development project.

- Module public API is exactly index.ts.
- No import path points into another module folder.
- All shared types come from contracts.
- All infrastructure comes from platform adapters.
- Module works using mock data.
- Loading/error/offline/empty states are covered where relevant.
- Tests pass.
- README states purpose, files, I/O, run/test commands and limitations.
- Mounting the module requires only registry/route/dependency wiring.
- No hidden side effects or undocumented events exist.

Build order

Stage	Work	Exit condition
1	Contracts + App Shell	Compile and route skeleton works.
2	Simulator	Standard snapshots/events can be produced deterministically.
3	Onboarding	Simulated Robo can be added.
4	Home + Energy	Dashboard uses contracts only.
5	Control + Safety	Commands are safety-gated; faults testable.
6	Talk	Mock AI chat works without hardware.
7	Camera + Environment	Independent sensor/camera UI works.
8	Activity + Health + Notifications	Events/diagnostics visible.
9	Analytics + Settings	History and preferences work.
10	Real adapters	Swap mock adapters for real implementations.

Real-hardware principle

DEVELOPMENT
Module → Adapter Interface → Mock Adapter → Simulator

PRODUCTION
Module → Adapter Interface → ESP32/Camera/AI Adapter → Device

The module contracts do not change during this swap.

23. Final Engineering Rules

Rule 1 — Scope is a technical requirement. Extra code is not automatically better code.

Rule 2 — Interfaces before integrations. Define the contract first, then implement mock, then production adapter.

Rule 3 — Modules own their internals. Another module may consume the output contract, never the private store/service.

Rule 4 — Safety is deterministic. A language model can explain/recommend, but physical commands must pass deterministic validation.

Rule 5 — Simulator first. Hardware absence must not block UI/module development.

Rule 6 — Evidence over claims. Analytics must label missing/incomplete data instead of inventing numbers.

Rule 7 — If the model is unsure, stop and report. Do not silently expand scope.

END OF SPECIFICATION